Practica 3 Proyectos integrados: Docker
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Exercicio 1:
Terminal principal

```
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker run -it alpine
/ # exit
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps -a
CONTAINER ID   IMAGE               COMMAND     CREATED          STATUS                      PORTS     NAMES
7fe01b150b87   alpine              "/bin/sh"   15 seconds ago   Exited (0) 10 seconds ago             naughty_wiles
bf2f5af0ef5f   hello-world:latest  "/hello"    42 minutes ago   Exited (0) 42 minutes ago             clever_chebyshev
fed95088e6c8   hello-world         "/hello"    2 hours ago      Exited (0) 2 hours ago                great_leakey
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker start naughty_wiles
naughty_wiles
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker attach naughty_wiles
/ # ^C

/ # esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps -a
CONTAINER ID   IMAGE               COMMAND     CREATED          STATUS                        PORTS     NAMES
7fe01b150b87   alpine              "/bin/sh"   8 minutes ago    Exited (137) 27 seconds ago           naughty_wiles
bf2f5af0ef5f   hello-world:latest  "/hello"    51 minutes ago   Exited (0) 51 minutes ago             clever_chebyshev
fed95088e6c8   hello-world         "/hello"    2 hours ago      Exited (0) 2 hours ago                great_leakey
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker rm naughty_wiles
naughty_wiles
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps.a
docker: unknown command: docker ps.a

Run 'docker --help' for more information
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps -a
CONTAINER ID   IMAGE               COMMAND     CREATED          STATUS                      PORTS     NAMES
bf2f5af0ef5f   hello-world:latest  "/hello"    51 minutes ago   Exited (0) 51 minutes ago             clever_chebyshev
fed95088e6c8   hello-world         "/hello"    2 hours ago      Exited (0) 2 hours ago                great_leakey
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker run -it --rm alpine
/ # exit
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker rmi alpine:latest
Untagged: alpine:latest
Deleted: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker images
                                                                                                  i Info →   U  In Use
IMAGE               ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest  ef54e839ef54       25.9kB         9.52kB    U
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $
```

Terminal Secundaria:

```
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker stop naughty_wile
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps -a
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ newgrp docker
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker stop naughty_wiles
naughty_wiles
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps -a
CONTAINER ID   IMAGE               COMMAND     CREATED          STATUS                        PORTS     NAMES
7fe01b150b87   alpine              "/bin/sh"   8 minutes ago    Exited (137) 23 seconds ago           naughty_wiles
bf2f5af0ef5f   hello-world:latest  "/hello"    50 minutes ago   Exited (0) 50 minutes ago             clever_chebyshev
fed95088e6c8   hello-world         "/hello"    2 hours ago      Exited (0) 2 hours ago                great_leakey
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps
CONTAINER ID   IMAGE     COMMAND     CREATED          STATUS          PORTS     NAMES
e16b14b1cc0d   alpine    "/bin/sh"   12 seconds ago   Up 12 seconds             suspicious_swirles
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps -a
CONTAINER ID   IMAGE               COMMAND     CREATED          STATUS                      PORTS     NAMES
bf2f5af0ef5f   hello-world:latest  "/hello"    54 minutes ago   Exited (0) 54 minutes ago             clever_chebyshev
fed95088e6c8   hello-world         "/hello"    2 hours ago      Exited (0) 2 hours ago                great_leakey
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
esmerodes@esmerodes-ASUSLaptop-TN3402YA: $
```

Explicación do ex 1

Nesta práctica, realizamos unha serie de operacións co contedor Alpine para entender o ciclo de vida dos contedores Docker: creación, execución, detención, reinicio, eliminación e o efecto da opción --rm.

1. **Executamos un contedor Alpine sen a opción --rm**
Comando:

`docker run -it alpine`

Usamos -it para executar o contedor de forma interactiva (-i mantén STDIN aberto e -t asigna un terminal pseudo), o que nos permite acceder ao shell /bin/ash de Alpine. Sen --rm, o contedor non se elimina automaticamente ao saír, quedando detido no sistema para poder xestionarlo despois.

2. **Saímos do contedor**
Comando dentro do contenedor:

`exit`

Ao saír co exit, o proceso principal (o shell) termina → o contedor pasa a estado Exited (detido), pero segue existindo porque non usamos --rm. Isto demostra que os contedores son efémeros só se se configura así.

3. **Comprobamos o estado do contedor**
Comando:

`docker ps -a`
docker ps só mostra contedores en execución (Up). Con -a (all) vemos tamén os detidos (Exited). Aquí vemos o noso contedor con status Exited (0), nome aleatorio (ex: festive_pascal) e ID.

4. **Reiniciamos o contedor sen abrir o terminal interactivo**
Comando:

`docker start <nome_ou_ID>` (ex: `docker start festive_pascal`)

docker start reinicia un contedor detido usando o mesmo comando orixinal da imaxe (neste caso /bin/ash). Sen -i, non se abre terminal interactivo → o contedor iníciase e remata case inmediatamente (porque o shell sen input termina). Vemos que pasa a Exited de novo.

5. **Accedemos ao terminal interactivo do contedor**
Comando:

`docker start -i <nome_ou_ID>`

A opción -i fai que o contedor sexa interactivo de novo → nos mete directamente no shell / #. Isto é útil para contedores que necesitan entrada humana.

6. **Desde outro terminal, detemos o contedor e comprobamos o efecto na sesión interactiva**

Comando (noutro terminal): **`docker stop <nome_ou_ID>`**

docker stop envía SIGTERM ao proceso principal (o shell) → a sesión interactiva na terminal principal péchase automaticamente (perdemos o prompt e volvemos ao host). Isto demostra o illamento: detendo dende fóra afecta ao proceso dentro.

7. Eliminamos o contedor e comprobamos

Comando: **`docker rm <nome_ou_ID>`**

docker rm elimina o contenedor detido. Despois de eliminar, xa non aparece en docker ps -a.

8. Ex**ecutamos un novo contedor Alpine con --rm**

Comando: **`docker run -it --rm alpine`**

A opción `--rm` fai que o contedor se elimine automaticamente ao rematar (ao saír).

9. **Desde outro terminal, comprobamos que está activo**

Comando: **`docker ps`**

Mentres estamos dentro, o contedor está Up (en execución).

10. **Saímos do contedor e comprobamos que non aparece**
Comando:
`exit`

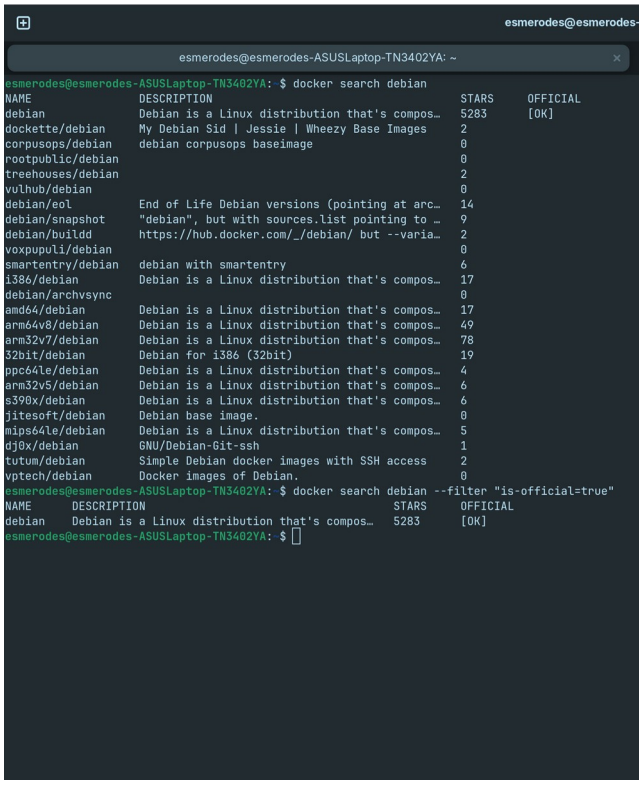
Ao saír, o contedor remata e --rm elimínao automaticamente → non aparece nin en `docker ps` nin en `docker ps -a`.

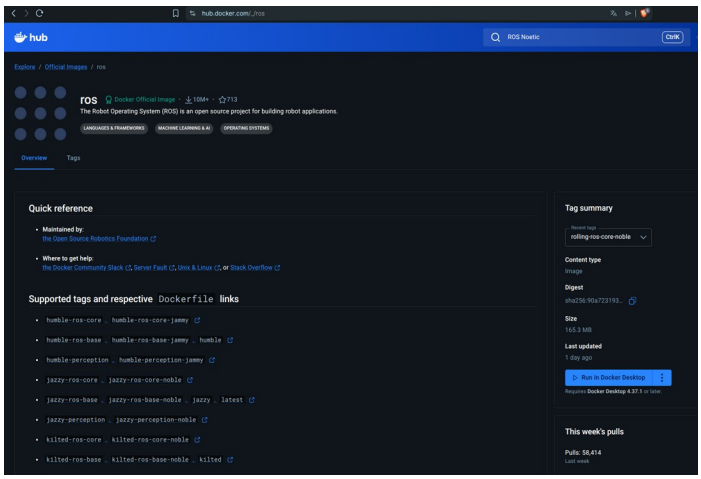
11. **Eliminamos a imaxe e comprobamos**

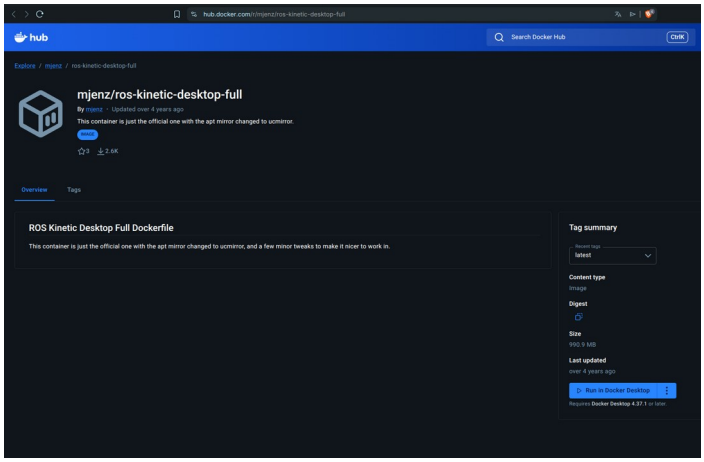
Comando: **`docker rmi alpine`**

`docker rmi` elimina a imaxe local. Só se pode facer se non hai contedores dependentes dela. Despois, `docker images` xa non a mostra.

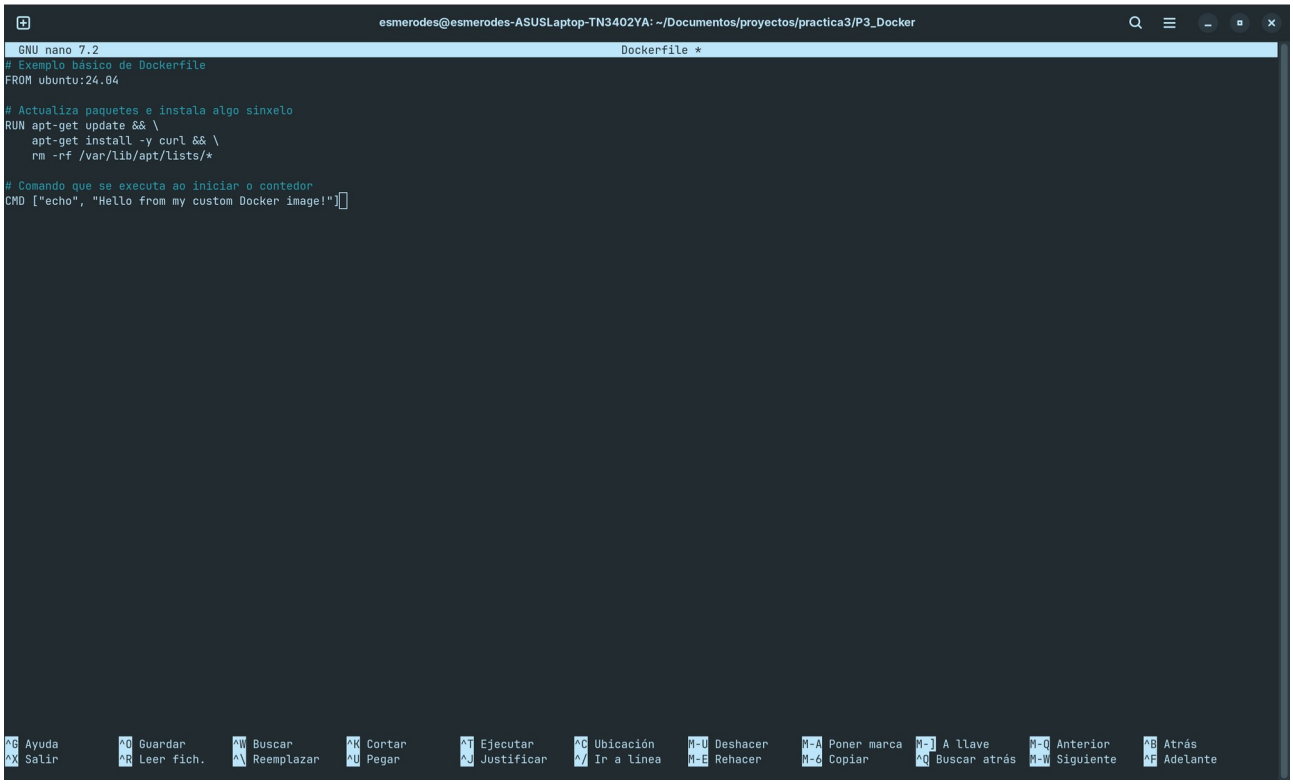
Execicio 2:







Exercicio 3:



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```
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~/Documentos/proyectos/practica3/P3_Docker

esmerodes@esmerodes-ASUSLaptop-TN3402YA: $ cd /home/esmerodes/Documentos/proyectos/practica3
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3$ mkdir P3_Docker
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3$ dc P3_Docker
dc: Will not attempt to process directory P3_Docker
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3$ cd /home/esmerodes/Documentos/proyectos/practica3/P3_Docker
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker$ nano Dockerfile
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker$ docker build -t minha-imaxe:1.0 .
[+] Building 13.7s (6/6) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                            0.0s
 => => transferring dockerfile: 322B                                                                                            0.0s
 => [internal] load metadata for docker.io/library/ubuntu:24.04                                                                 1.9s
 => [internal] load .dockerignore                                                                                               0.0s
 => => transferring context: 2B                                                                                                 0.0s
 => [1/2] FROM docker.io/library/ubuntu:24.04@sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9            2.4s
 => => resolve docker.io/library/ubuntu:24.04@sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9            0.0s
 => => sha256:01d7766a2e4a62b74e0bebf2cd12c47e675e9221174f6570854203e84ffe68b0 29.73MB / 29.73MB                                1.9s
 => => extracting sha256:01d7766a2e4a62b74e0bebf2cd12c47e675e9221174f6570854203e84ffe68b0                                       0.4s
 => [2/2] RUN apt-get update &&     apt-get install -y curl &&     rm -rf /var/lib/apt/lists/*                                 8.8s
 => exporting to image                                                                                                          0.5s
 => => exporting layers                                                                                                         0.3s
 => => exporting manifest sha256:aa69bb7398f0749b3f5b50989b0ed1befe192bef0a1cb158acfab57abeb46581                               0.0s
 => => exporting config sha256:e4faa81ca558454faef600f1580562e7f5e55b15773e820cb014d72e1a5c8167                                 0.0s
 => => exporting attestation manifest sha256:fa5b3e4f765baa6b39e000277c7a0fbaf753457295df5f64fedfbe0768b22525                   0.0s
 => => exporting manifest list sha256:65e140fcf15957574a6b518b2937d50706c81d8e43242e6b733cdfe90e2518c2                          0.0s
 => => naming to docker.io/library/minha-imaxe:1.0                                                                              0.0s
 => => unpacking to docker.io/library/minha-imaxe:1.0                                                                           0.1s
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker$ docker images
                                                                                                   i Info →   U  In Use
IMAGE               ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest  ef54e839ef54        25.9kB         9.52kB    U
minha-imaxe:1.0     65e140fcf159         132MB         33.4MB
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker$ docker run minha-imaxe:1.0
Hello from my custom Docker image!
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker$ docker rmi minha-imaxe:1.0
Error response from daemon: conflict: unable to delete minha-imaxe:1.0 (must be forced) - container fdf4213cb61c is using its referenced image 65e140fcf159
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker$
```

Explicacion ex3:

Primerio accedo a carpeta que teño asociada a asignatura, creo a carpeta con mkdir P3_Dockers y accedo a ella con cd.
Creo el archivo con una instruciones, FROM para partir dunha base oficial (Ubuntu 24.04), RUN para instalar dependencias durante a construción (curl como exemplo), e CMD para definir o proceso default ao executar o contedor. Isto crea unha imaxe personalizada reproducible.

Construímos a imaxe a partir do Dockerfile
Comando: docker build -t minha-imaxe:1.0 .docker build le o Dockerfile e executa as instrucións paso a paso, creando capas. O tag -t nomea a imaxe (minha-imaxe:1.0). O punto . indica o contexto actual. A saída mostra cada step e o ID final da imaxe.

Comprobamos a imaxe creada
Comando: docker images
Lista todas as imaxes locais.
Vemos a nosa minha-imaxe:1.0 co tamaño e data de creación.
Executamos un contedor desde a imaxe
Comando: docker run minha-imaxe:1.0
Crea e executa un contedor da imaxe, correndo o CMD definido → imprime a mensaxe. Demostra que a imaxe funciona correctamente.

Exercicio 4

```
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~/Documentos/proyectos/practica3/P3_Docker/ex4

esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ nano run.sh
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ cat run.sh
#!/bin/sh
ping -c 5 $@
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ nano Dockerfile
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ cat Dockfile
cat: Dockfile: No existe el archivo o el directorio
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ cat Dockerfile
FROM alpine:latest

# Copia o script ao raiz e dá permisos
COPY run.sh /
RUN chmod +x /run.sh

# Define ENTRYPOINT (comando fixo que se executa sempre)
ENTRYPOINT ["/run.sh"]

# Define CMD (parámetro default para ENTRYPOINT)
CMD ["8.8.8.8"]

esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ docker build -t ping:v1 .
[+] Building 3.1s (8/8) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 284B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/alpine:latest                                                              1.8s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 56B                                                                                              0.0s
 => [1/3] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659        0.8s
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659          0.0s
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB                                0.7s
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153                                     0.1s
 => [2/3] COPY run.sh /                                                                                                       0.0s
 => [3/3] RUN chmod +x /run.sh                                                                                                0.1s
 => exporting to image                                                                                                        0.1s
 => => exporting layers                                                                                                       0.1s
 => => exporting manifest sha256:d80648dd8ae140a84b8e553f52b07c576b9e2df93d4126e7f24d86554afcc6d8                             0.0s
 => => exporting config sha256:8716a8cc6fb4ed404bb2963970ea79e4290491d595316e7d061a7aa39ef5d76f                               0.0s
 => => exporting attestation manifest sha256:1952d5175644cbe93a1a41f3a9488514262df9dfa1c8978f761910067fe4ef1                  0.0s
 => => exporting manifest list sha256:2dba911c6b6464391361ee4195deaacd98b7a3586cc4ba4c309d10487e18fbf4                        0.0s
 => => naming to docker.io/library/ping:v1                                                                                    0.0s
 => => unpacking to docker.io/library/ping:v1                                                                                 0.0s
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ docker images
                                                                                                   i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   ef54e839ef54       25.9kB         9.52kB    U
minha-imaxe:1.0      65e140fcf159        132MB         33.4MB    U
ping:v1              2dba911c6b64         13MB         3.87MB
```

continue

```
esmerodes@esmerodes-ASUSLaptop-TN3402YA: ~/Documentos/proyectos/practica3/P3_Docker/ex4

 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 56B                                                                                              0.0s
 => [1/3] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659        0.8s
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659          0.0s
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB                                0.7s
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153                                     0.1s
 => [2/3] COPY run.sh /                                                                                                       0.0s
 => [3/3] RUN chmod +x /run.sh                                                                                                0.1s
 => exporting to image                                                                                                        0.1s
 => => exporting layers                                                                                                       0.1s
 => => exporting manifest sha256:d80648dd8ae140a84b8e553f52b07c576b9e2df93d4126e7f24d86554afcc6d8                             0.0s
 => => exporting config sha256:8716a8cc6fb4ed404bb2963970ea79e4290491d595316e7d061a7aa39ef5d76f                               0.0s
 => => exporting attestation manifest sha256:1952d5175644cbe93a1a41f3a9488514262df9dfa1c8978f761910067fe4ef1                  0.0s
 => => exporting manifest list sha256:2dba911c6b6464391361ee4195deaacd98b7a3586cc4ba4c309d10487e18fbf4                        0.0s
 => => naming to docker.io/library/ping:v1                                                                                    0.0s
 => => unpacking to docker.io/library/ping:v1                                                                                 0.0s
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ docker images
                                                                                                   i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   ef54e839ef54       25.9kB         9.52kB    U
minha-imaxe:1.0      65e140fcf159        132MB         33.4MB    U
ping:v1              2dba911c6b64         13MB         3.87MB
ubuntu:24.04         d1e2e92c075e        119MB         31.7MB
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ docker run --rm ping:v1
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: seq=0 ttl=114 time=44.191 ms
64 bytes from 8.8.8.8: seq=1 ttl=114 time=31.650 ms
64 bytes from 8.8.8.8: seq=2 ttl=114 time=43.880 ms
64 bytes from 8.8.8.8: seq=3 ttl=114 time=43.333 ms
64 bytes from 8.8.8.8: seq=4 ttl=114 time=30.145 ms

--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 30.145/38.639/44.191 ms
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$ docker run --rm ping:v1 1.1.1.1
PING 1.1.1.1 (1.1.1.1): 56 data bytes
64 bytes from 1.1.1.1: seq=0 ttl=53 time=30.874 ms
64 bytes from 1.1.1.1: seq=1 ttl=53 time=45.679 ms
64 bytes from 1.1.1.1: seq=2 ttl=53 time=33.847 ms
64 bytes from 1.1.1.1: seq=3 ttl=53 time=45.504 ms
64 bytes from 1.1.1.1: seq=4 ttl=53 time=30.505 ms

--- 1.1.1.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 30.505/37.281/45.679 ms
esmerodes@esmerodes-ASUSLaptop-TN3402YA:~/Documentos/proyectos/practica3/P3_Docker/ex4$
```

Crea o script bash run.sh

Comando: nano run.sh

Pego este contido exacto:
```
#!/bin/sh
ping -c 5 $@
```
Crea un script que executa `ping -c 5` (5 paquetes) co argumento pasado ($@ = todos os argumentos, como a IP).

Crea o Dockerfile coas características pedidas
Comando:
```
nano Dockerfile
```
Pega este contido:
```
FROM alpine:latest

# Copia o script ao raíz e dá permisos
COPY run.sh /
RUN chmod +x /run.sh

# Define ENTRYPOINT (comando fixo que se executa sempre)
ENTRYPOINT ["/run.sh"]

# Define CMD (parámetro default para ENTRYPOINT)
CMD ["8.8.8.8"]
```
Que fai cada parte?
- `FROM alpine:latest`: Base lixeira (Alpine, ~5MB).
- `COPY run.sh /`: Copia o script do host ao raíz do contedor.
- `RUN chmod +x /run.sh`: Dá permisos de execución durante a construción.
- `ENTRYPOINT ["/run.sh"]`: Define o comando principal (fixo, non sobrescribible).
- `CMD ["8.8.8.8"]`: Parámetro default (IP de Google DNS para ping se non pasas outro).

Ao run, executa `/run.sh 8.8.8.8` por defecto, ou o que pases como argumento.

4. **Constrúe a imaxe co nome ping:v1**
Comando:
```bash
docker build -t ping:v1 .
```
Que fai? Constrúe a imaxe usando o Dockerfile, tag "ping:v1". Saída: steps con descarga de alpine, copy, chmod → Successfully tagged ping:v1.
Comproba: `docker images` → ves ping v1.

5. **Lanza un contedor para comprobar o default (8.8.8.8)**
Comando:
```bash
docker run --rm ping:v1
```
Que fai? Executa o contedor co ENTRYPOINT + CMD default. Saída: ping -c 5 8.8.8.8 (mostra 5 paquetes enviados/recibidos).

--rm: elimínao automaticamente para non deixar lixo.

6. **Lanza un novo contedor con IP diferente (ex. 1.1.1.1)**
   Comando:
   ```bash
   docker run --rm ping:v1 1.1.1.1
   ```
   Que fai? Sobrescribe o CMD co argumento "1.1.1.1" (IP de Cloudflare DNS). Saída: ping -c 5 1.1.1.1.
   Comproba que cambia a IP correctamente.